The Crime & Society Review

Rethinking Crime, Justice & Society
Emerging Perspectives for the 21st Century
Volume I ; Issue I (Oct-Dec/2026)
Online/Fortnightly/Multi-Disciplinary/English/2026

Expectancy Calculations, Hygiene Expectations, and Task Productivity: A Quantitative Appraisal of Salary Adjustments in Selected Nigerian Pharmaceutical Manufacturing Enterprises

Acceptance: Nov 02, 2026 ; Publication: Nov 10, 2026

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Abstract

The above study is a quantitative analysis of the influence of employees' perception of salary increase policies on the motivational status and productivity of operations in two major pharmaceutical industries in Osun State, Nigeria namely: De-Shalom Pharmaceutical Laboratory Nigeria Limited (Ilesa) and Sam-Ace Limited (Ede). The theoretical framework adopted for the study was based on the expectancy theory by Victor Vroom (M=E×I×V) and Two-factor theory by Frederick Herzberg. The sample used in the study comprises of n=367 out of N=4,501 members selected by Taro Yamane method of formula and stratified random sampling in production, quality control, administration, and security departments. Quantitative data were gathered through a structured questionnaire with a four-point Likert scale using SPSS software version 28. Results show that salary is viewed as an important hygiene factor, with its absence leading to extreme dissatisfaction and demotivation among workers as 53.6% of the respondent's report lack of procedural clarity and fairness in current wage adjustment ($\bar{x}$=1.85). Nevertheless, when salary increments are seen as possible rewards tied to performance, there is a huge increase in expectations and instrumentality perceptions, leading to increased levels of work effort (80.3%), job dedication (62.9%) and daily task effectiveness (72.7%). The study concludes that salary increments should be implemented in a manner that is transparent as motivation rewards based on merit and not just maintenance rewards arbitrarily.

Keywords: Salary Increase, Expectancy Theory, Herzberg Two-Factor Theory, Employee Motivation, Operational Productivity, Hygiene Factors, Pharmaceutical Industry.

1. Introduction

Organizational management of employee compensation continues to be one of the core pillars of organizational success, talent retention, and labor relations in the current environment of developing nations (Abbah, 2014; Adeyemi & Ojo, 2024). In the case of Nigeria, macroeconomic instability, high rates of inflation, and devaluation of currency have continued to reduce the purchasing power of the populace, thus making salary management an important aspect of organizational assessment (Adanlawo

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& Nkomo, 2023; Dajo & Akor, 2022). When salary structures do not reflect economic realities or are opaque in nature, organizations typically end up having low morale and high labor turnover.

Pharmaceutical Manufacturing Industry in South-Western Nigeria, represented by companies like De-Shalom Pharmaceutical Laboratory Nigeria Limited (in Ilesa, Osun State) and Sam-Ace Limited (Gbongan-Ibadan Road, Ede, Osun State), is characterized by a very strict regulatory framework, which involves accuracy, constant monitoring, and adherence to GMP (Good Manufacturing Practices) (Alkandi et al., 2023). Adherence to high standards is mostly dependent on motivation, dedication, and attention of people working on factory floors, quality control specialists, and administrative staff (Alkandi et al., 2023). However, the process of wage evaluation in semi-urban manufacturing plants tends to be perceived as opaque and inconsistent (Ali, 2020).

Although many past studies have considered the compensation structure, few empirical studies have examined salary increments through the combined perspective of Vroom's Expectancy Theory and Herzberg's Motivation-Hygiene Theory in the Nigerian pharmaceutical manufacturing industry (Sowunmi, 2024). The current study fills this void by exploring how pharmaceutical employees appraise the process of salary increments, compute the expectancy-instrumentality drivers and hygiene expectations, and the effects of these expectations on productivity and efficiency of tasks performed each day.

1.1 Research Questions

1. What are the perceptions of employees regarding salary increase practices in De-Shalom Pharmaceutical Laboratory Nigeria Limited and Sam-Ace Limited?
2. How does salary increase influence employee motivation through the dual lens of expectancy calculations and motivator-hygiene dynamics?
3. In what ways do salary increments impact operational productivity and task efficiency within the selected pharmaceutical manufacturing firms?

2. Review Of Literature & Theoretical Framework

2.1 Conceptual Review

Salary Adjustments and Compensation Governance

The term salary increase can be defined as any increment made on the base monetary pay taking into account performance appraisals, experience, cost of living, or market-based increments (Alananzeh et al., 2023; Adeyemi & Ojo, 2024). The pay governance is the system of regulations, appraisal, and communication that guide pay adjustments (John, 2024). In manufacturing environments, a good pay governance system makes sure that employees understand the relationship between their performance and rewards (Paananen, 2024).

Workplace Motivation

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Motivation is referred to as the psychological forces that drive and give direction and energy to the behavior of individuals for accomplishing organizational objectives (Ryan & Deci, 2023). Motivation is divided into intrinsic factors (getting satisfaction out of the activity itself) and extrinsic factors (involving external monetary and non-monetary rewards) (Zhou & Zhang, 2024). In manufacturing plants, extrinsic monetary motivators play a critical role in ensuring individual efforts match organizational production objectives (Alkandi et al., 2023).

Operational Productivity

Motivation is referred to as the psychological forces that drive and give direction and energy to the behavior of individuals for accomplishing organizational objectives (Ryan & Deci, 2023). Motivation is divided into intrinsic factors (getting satisfaction out of the activity itself) and extrinsic factors (involving external monetary and non-monetary rewards) (Zhou & Zhang, 2024). In manufacturing plants, extrinsic monetary motivators play a critical role in ensuring individual efforts match organizational production objectives (Alkandi et al., 2023).

2.2 Theoretical Framework

This research incorporates both major motivational process/content theories: Expectancy Theory (V. Vroom, 1964) and Two-Factor Theory (F. Herzberg, 1959).

1. Victor Vroom's Expectancy Theory

According to Vroom (1964), the effort a person puts in (M) depends on the product of three major components through the following cognitive calculation:

$$M = E \times I \times V$$

Expectancy (E→P): Expectancy refers to the subjective probability that the effort (E) put forward by the individual will lead to task performance (P).

Instrumentality (P→R): Instrumentality is the subjective probability that the task performance (P) will lead to organizational reward (R) such as salary raise.

Valence (V): Valence refers to the value that an individual attaches to organizational reward (R).

If any one of the variables is zero, for instance when workers perceive that performance does not guarantee increment in salary (I=0) or when workers are paid without considering their performance (E=0), motivation will become zero irrespective of the worth of the incentive (V).

2. Frederick Herzberg's Two-Factor Theory

Herzberg (1959) separated work-related variables into two independent categories:

- **Hygiene Variables (Extrinsic / Dissatisfiers)**: Such variables as base pay, organizational policies, supervision, and physical work environment. Satisfactory hygiene variables prevent job dissatisfaction, but do not generate intrinsic job satisfaction or motivation.

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- **Motivators (Intrinsic / Satisfiers)**: Such variables as achievement, recognition, responsibility, and personal development. They lead to job satisfaction and increased effort.

In the context of compensation management, base pay is a hygiene variable, and unsatisfactory or inequitable pay leads to serious dissatisfaction. In contrast, if increases in salary are structured in a way that they become performance-based, then they acquire features of a motivator by offering recognition of performance (Herzberg, 1968; Kaufman, 2023).

2.3 Integrated Theoretical Model Narrative

- **Hygiene Threshold (Herzberg):** Basic salary and the cost-of-living increase meet the requirements of hygiene, ensuring that there is no dissatisfaction and administrative frustration at work.
- **Cognitive Process (Vroom):** The employee judges the performance appraisal process. In case the employee believes in high expectancy (E: "My efforts will result in good performance") and high instrumentality (I: "Good performance will result in a decent salary raise") along with high valence (V: "Salary raise is important to me"), then motivation (M) is greatly increased.
- **Motivator (Herzberg):** Performance-based salary raise is acknowledged as an explicit recognition of achievement and importance, thereby turning compensation into a motivator.
- **Consequences of behavior:** It is the double process of psychology that makes people work harder and perform better.

3. Methodology

3.1 Research Design & Study Area

The descriptive survey research design was used in this study to obtain quantitative data from two large industrial pharmaceutical organizations within Osun State, Nigeria:

De-Shalom Pharmaceutical Laboratory Nigeria Limited: Peace Haven, Oke Ooye Street, Iyemogun, Ilesa, Osun State (specialized in compounding, oral liquids, and tablets production).

Sam-Ace Limited: Km 14 Gbongan-Ibadan Road, Ede, Osun State (specialized in OTC pain relievers, ORS, and liquids).

3.2 Target Population and Sample Size

The target population size (N): N=4,501 employees from both companies (2,201 for De-Shalom and 2,300 for Sam-Ace) including Production, Quality Control, Human Resources/ Administration, and Security/Maintenance.

The sample size (n): Based on the Taro Yamane equation with a 0.05 margin of error:

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$$n = \frac{N}{1+N}$$

3.3 Sampling & Data Collection

Stratified random sampling method was employed for the distribution of structured questionnaires (EPSIMPQ). The tool was developed using a 4-point Likert scale (where 4 indicates "Strongly Agree [SA]", 3 indicates "Agree [A]", 2 indicates "Disagree [D]", 1 indicates "Strongly Disagree [SD]"). 100% retrieval rate (n=367) was obtained.

3.4 Data Analysis

The data analysis was carried out using SPSS version 28. The descriptive statistics such as Frequencies, Percentages, Means [$\bar{x}$], and Standard Deviations [SD] were used. A mean criterion of $\bar{x} \geq 2.50$ was set for agreement.

Sex	Male	200	54.5%
	Female	162	44.1%
	Prefer not to say	5	1.4%
Age Group	18 – 25 years	120	32.7%
	26 – 44 years	209	56.9%
	45 years and above	38	10.4%
Organization	De-Shalom Pharmaceutical Lab.	201	54.8%
	Sam-Ace Limited	166	45.2%
Highest Education	Primary School Certificate	24	6.5%
	Secondary School (SSCE)	48	13.1%
	OND / NCE Diploma	142	38.7%
	HND / B.Sc Degree	131	35.7%
	Postgraduate Degree	22	6.0%
Length of Service	Less than 1 year	56	15.3%
	1 – 3 years	179	48.8%
	4 – 6 years	101	27.5%
	7 – 10 years	27	7.4%
	Above 10 years	4	1.1%

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4. Results & Empirical Findings

4.1 Socio-Demographic Profile of Respondents (n = 367)

Table 1: Socio-Demographic Profile of Sampled Respondents

4.2 Research Question 1: Employees' Perceptions of Salary Increase Practices

Table 2: Employees' Perceptions of Salary Increase Practices and Governance

Perceptual Evaluation Statement	SA (4)	A (3)	D (2)	SD (1)	Mean ($\bar{x}$)	*SD*	Decision
Process Transparency: The salary increase process is fair and transparent.	40 (10.8%)	50 (13.6%)	35 (9.5%)	197 (53.6%)	1.85	1.09	Disagree
Criteria Clarity: I clearly understand criteria used for salary increments.	36 (9.8%)	33 (8.9%)	50 (13.6%)	208 (56.6%)	1.78	1.04	Disagree
Merit Linkage: Salary increases are genuinely based on performance and merit.	37 (10.0%)	80 (21.7%)	40 (10.8%)	200 (54.4%)	1.88	1.11	Disagree
Review Frequency: I am satisfied with the frequency of salary reviews.	22 (5.9%)	30 (8.2%)	78 (21.2%)	205 (55.8%)	1.63	0.93	Disagree

Source: Field Survey, 2024. Benchmark Decision Mean = 2.50.

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4.3 Research Question 2: Influence of Salary Increase on Employee Motivation

Table 3: Influence of Salary Increase on Motivation (Expectancy & Motivator Dimensions)

4.4 Research Question 3: Impact of Salary Increments on Employee Productivity

Table 4: Impact of Salary Increments on Operational Productivity and Output

Operational Impact Statement	SA (4)	A (3)	D (2)	SD (1)	Mean ($\bar{x}$)	*SD*	Decision
Productivity Drive: Salary increases encourage me to maintain high productivity.	295 (80.3%)	45 (12.2%)	8 (2.1%)	14 (3.8%)	3.69	0.73	Agree
Initiative & Extra Effort: I take more initiative after receiving a fair pay raise.	275 (74.9%)	68 (18.5%)	12 (3.2%)	3 (0.8%)	3.65	0.61	Agree
Target Compliance: A good salary review motivates me to meet production targets.	286 (77.9%)	48 (13.0%)	14 (3.8%)	10 (2.7%)	3.65	0.71	Agree
Task Focus & Efficiency: Salary increases improve daily operational efficiency.	267 (72.7%)	60 (16.3%)	5 (1.3%)	18 (4.9%)	3.54	0.86	Agree

Source: Field Survey, 2024. Benchmark Decision Mean = 2.50.

5. Discussion

Empirical findings shed light on the motivational pattern in the investigated pharmaceutical production companies. With regard to Research Question 1 (Table 2), it is noted that the majority of the employees are dissatisfied with the current pay review system. There was significant disagreement regarding pay transparency ($\bar{x}$=1.85) and pay for performance ($\bar{x}$=1.88). According to Herzberg's Two-Factor Theory, the salary in question first of all plays a role of the basic hygiene factors. As soon as the pay

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review process is seen as arbitrary or uncertain, serious dissatisfaction occurs (Alananzeh et al., 2023; Greenberg & Colquitt, 2023). From the perspective of Vroom's Expectancy Theory, such uncertainty lowers instrumentality (I→0); when employees doubt that increased efforts will lead to the raise, motivation falls down (Ali, 2020).

In contrast, findings from Research Questions 2 and 3 (Tables 3 and 4) clearly indicate that if the salary increase is seen as a fair, reasonable, and realistic process, it plays a very important role in influencing the behavior of employees. More than 80% of the respondents strongly agreed that fair salary increases result in increased productivity on a daily basis ($\bar{x}$=3.69), and about 69.4% said that pay increases have increased job satisfaction and morale ($\bar{x}$=3.50). This supports the concept of Vroom's model, whereby when employees see there is a clear path connecting effort to performance and performance to valued reward (E×I×V), the motivation to work increases. When pay increases are used as a form of reward for performance, salary goes beyond being a simple hygiene factor and serves as a motivator (Kaufman, 2023; Sowunmi, 2024).

6. Conclusion & Policy Recommendations

6.1 Conclusion

This study investigates how the employees perceive the process of salary increase, workplace motivation, and organizational productivity at De-Shalom Pharmaceutical Laboratory Nigeria Limited and Sam-Ace Limited based in Osun State, Nigeria. Drawing from the framework of the Expectancy Theory by Vroom and Two Factor Theory by Herzberg, it can be seen that although the existing process of wage review is considered opaque and inconsistent, salary is still a crucial motivating factor for the employees' work and productivity. The process of transparent salary payment is a source of hygiene factors and motivates the employees.

6.2 Policy Recommendations

- **Enhancing Instrumentality by Ensuring Good Pay Governance:** The management should define and communicate performance evaluation standards objectively, which have a direct relation between performance evaluations and salary increases.
- **Implementing Pay Review Cycles:** A consistent cycle of annual or biannual reviews should be put into place to ensure that wages keep up with inflation, retaining the role of salary as a good hygiene factor.
- **Improving Appraisal Communication:** HR managers should give structured feedback on performance evaluations to increase expectancy (E→P) of employees.

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